

THE TRANSITION FROM COMPUTER VISION TO LARGE VISUAL MODELS (LVMS): WHAT ARE THE IMPLICATIONS FOR ENTERPRISES AND VENDORS?

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Reece Hayden, Principal Analyst

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1. KEY FINDINGS

Although technologically mature, Computer Vision (CV) still offers plenty of commercial opportunities. But with the emergence of transformer-based models, ABI Research expects the vision Artificial Intelligence (AI) market to slowly begin to augment or replace deployments with Large Vision Models (LVMs) and Vision Language Models (VLMs). These models offer new capabilities and will eventually enable more complex use cases. However, the vision AI market will not shift as quickly as language-based models given the numerous technology and commercial challenges that enterprises and vendors face.

- Computer vision may be a mature technology, but commercial opportunities remain with existing customers still looking to scale deployments and plenty of long-tail enterprise unexplored.
- LVMs are emerging, but they are still in search of a widely applicable and value-added use case. Without use case-driven demand, the market will continue to focus on computer vision for the short to medium term. However, certain market segments and regions will adopt LVMs faster than others—China is already investing here, especially for automotive.
- Hybrid AI systems that integrate CV and LVMs across cloud, edge, and devices will be the entry point for early LVM adopters.
- Lightweight domain-specific vision models will be an enabler supporting the development of the LVM market. Optimization and data are key areas to focus on.
- Early LVM adoption will be cloud-based, targeting low-risk, low-performance use cases supporting new experiences or analytics.
- Although foundation models are emerging, the LVM and VLM markets are commercially and technologically underdeveloped: lacking core ecosystem partners, e.g., System Integrators (SIs) and targeted Machine Learning Operations (MLOps) solutions. Before any growth, this ecosystem needs to be built out.
- Long-term market dynamics will push toward physical AI agents based on the deep integration of LVMs, digital twins, and robotics.

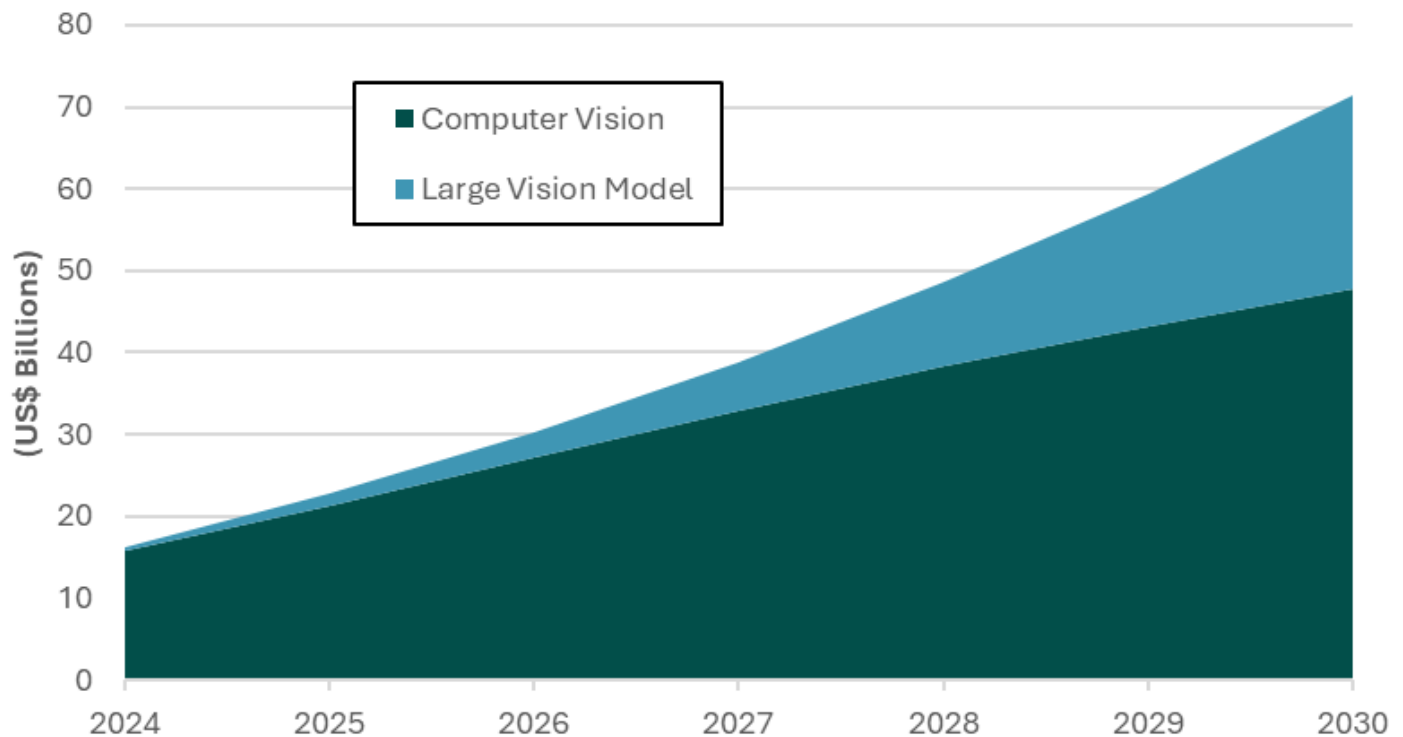
2. KEY FORECASTS

Enterprises continue to invest in CV; however, over the next 6 years, some of this spending will transition to LVMs with a Compound Annual Growth Rate (CAGR) of 95% with spending focused on data, optimization, and domain-specific training.

Chart 1: Spending on Vision AI Software

World Markets: 2024 to 2030

(Source: ABI Research)

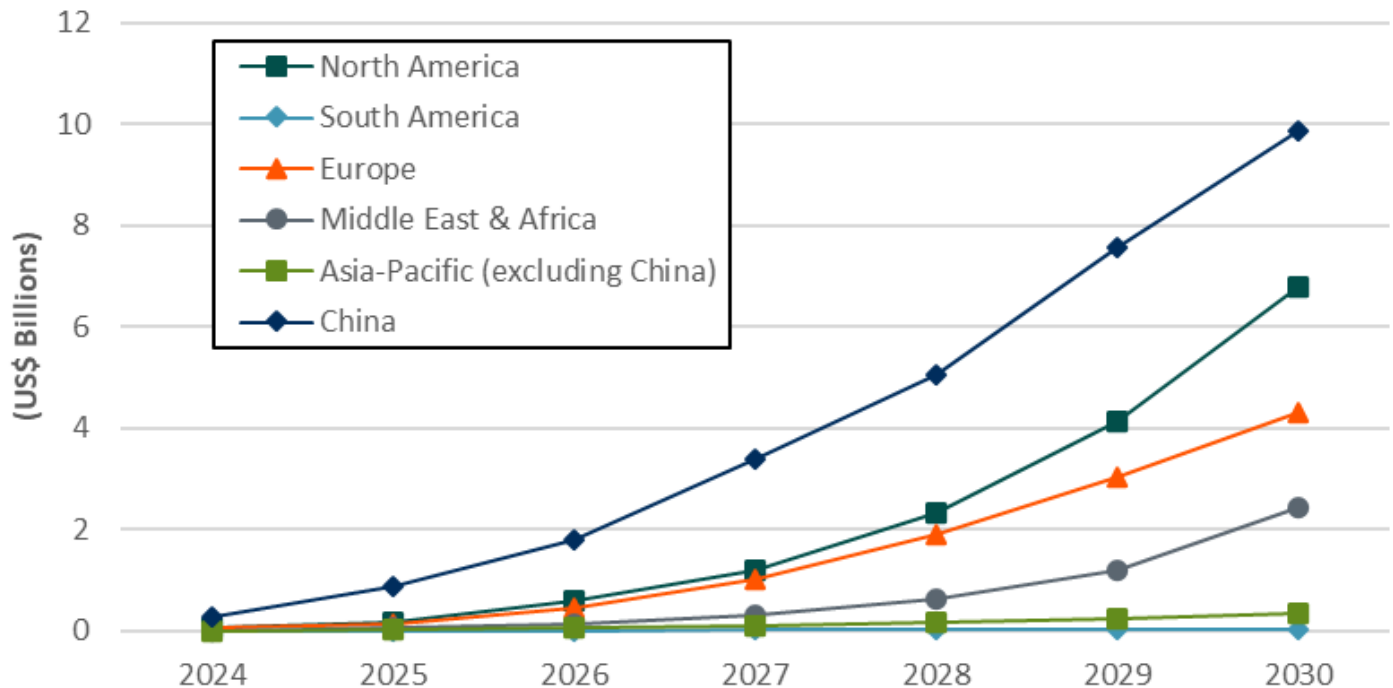


The earliest adopters of LVMs have been China supported by its rapidly developing autonomous vehicle and robotics markets. This early leadership will be sustained moving forward with a CAGR of 83%.

Chart 2: Regional Breakdown of LVM Software Spending

2024 to 2030

(Source: ABI Research)



3. KEY COMPANIES AND ECOSYSTEMS

Unlike the more mature language-based AI market, vision AI has not gone through much consolidation, due to entailing a far more fragmented market space with lots of vendors providing isolated tools/solutions.

3.1. ROBOFLOW

Roboflow provides a comprehensive platform for building, training, and deploying CV models. Roboflow facilitates the use and fine-tuning of LVMs by offering tools for dataset management, model training, and deployment. Roboflow introduced Autodistill, an open-source tool that uses large, foundation models to label data for training smaller, faster supervised models, and Roboflow Inference, a production-ready inference server for CV, supporting deployment of various model architectures. It recently partnered with Flex Logic to optimize its inference targeted hardware for CV deployments.

3.2. SNORKEL AI

Snorkel AI specializes in programmatic data labeling and development, enabling organizations to efficiently create high-quality datasets for training Machine Learning (ML) models, including LVMs. This approach accelerates the development of LVMs by providing scalable and adaptable data pipelines. Snorkel aims to integrate itself across various platforms, including Amazon Web Services (AWS), recently joining its Independent Software Vendor (ISV) accelerator program.

3.3. LANDINGAI

LandingAI leverages LVMs to power its LandingLens platform for advanced visual AI applications. It focuses on a data-centric AI approach, prioritizing high-quality data to train and deploy robust LVMs for various CV tasks. LandingAI also develops domain-specific LVMs, tailoring model capabilities to specific industries such as manufacturing and healthcare. Its recent rebranding as a Vision AI platform highlights its commitment to blending CV and LVM deployment. LandingAI is accelerating its commercial strategy through strategic investor-partnerships, including FPT, an SI based in Vietnam. This comes alongside its partnership with Snowflake.

3.4. EDGE IMPULSE

Edge Impulse specializes in enabling the development and deployment of ML models on edge devices. It supports LVMs by integrating with NVIDIA's TAO Toolkit, allowing users to access a library of pre-trained models optimized for edge deployment. This integration facilitates the training of sophisticated vision models on NVIDIA Graphics Processing Units (GPUs) and their deployment on resource-constrained devices like microcontrollers and low-power AI accelerators. Building on its partnership with STMicroelectronics, it co-released a new microcontroller supported by Edge Impulse's platform.

4. ENTERPRISE LVM ADOPTION

Although plenty of commercial opportunities remain in the computer vision market, the technology is starting to hit its use case limitations, making it vital for vendors to explore LVMs in the enterprise space.

Table 1: Vertical Deployments for LVMs

(Source: ABI Research)

Vertical	Company	Computer Vision Application	Pain point	LVM Deployment
Entertainment	Netflix	Audience analytics	Audience engagement	Personalized recommendations
Ubisoft	Interactive story telling	Dynamic storytelling complexity	Enhanced interactive story telling	
Marketing	Coca-Cola	Visual sentiment analysis	Accuracy	Enhanced visual sentiment analysis
PepsiCo	Brand sentiment analysis	Market competition	Real-time market trend analysis	
Retail	Starbucks	Customer behavior analysis	Customer engagement	Personalized marketing
Nike	Inventory management	Low stock	Demand forecasting	
Manufacturing	BMW	Assembly line inspection	Production defects	Supply chain optimization

Even though some enterprises are using LVMs, early use cases have similar characteristics: non-mission-critical, non-latency-sensitive, and not highly distributed applications, which highlights the limitations of the technology and, importantly, enterprise trust. Although there are plenty of factors that will accelerate enterprise LVM adoption, adjacent technologies such as Augmented Reality (AR)/Virtual Reality (VR) will play a core role as their synergistic value proposition will create use cases necessitating LVM deployment.

Table 2: How Do LVMs Fit with Adjacent Technologies?

(Source: ABI Research)

Adjacent Technology	Synergistic Value Proposition	Expectation
Robotics	Enhance ability to sense and respond to the real world in unstructured environments; supports human-robot interactions; improve perception and decision making.	As processing power increases and LVMs become more “lightweight,” expect integration supported by digital twin-enabled training data.
AI Devices	Support the development of immersive applications & new experiences.	Challenging to implement LVMs as compute & power constrained. But will be supported through hybrid systems and light-weight models.
Autonomous Vehicles	Play critical role in enabling self-driving cars by being domain-agnostic and responding to new environments.	Developers lean heavily on LVMs to support model development; however, local resources will be a challenge given processing requirements.

5. SOLVING VISION AI CHALLENGES

Scaling CV and implementing LVMs pose technological challenges for the market. MLOps investment and Research and Development (R&D) will be necessary to address these issues.

5.1. DOMAIN-SPECIFIC, LIGHTWEIGHT LVMS

Investing in domain-specific, lightweight LVM models can help overcome the issues of cost, power consumption, heating, and added latency

LVMs, compared to CV models, are much more challenging to deploy, as they are larger and require additional compute and energy. This not only adds cost, creates power consumption issues, and increases heating problems, but most importantly, it adds latency and hinders deployment for “real-time” use cases. Solving these problems requires investment in domain-specific, lightweight models. Neural Architectural Search (NAS), which automates the development of a new model to solve specific problems and specific tasks, has started to create value in the CV market with Deci.ai (recently acquired by NVIDIA) using it to develop Yolo-NAS (a leading vision model). In addition, other developments of new optimization techniques will be necessary. Arcee AI is exploring model merging and spectrum techniques that aim to integrate model capabilities into single models.

5.2. SOLVING THE DATA PROBLEM

Tools that enable integrating real-world, expert labeled data with synthetic data will solve some of the accuracy issues LVMs can experience

The availability of good quality, structured data is limited, which inhibits model training. This is particularly problematic for LVMs, given their appetite for data. Synthetic data generation using digital twins is showing early value. This involves creating a virtual replica of a physical asset or system that uses real-time data to simulate its behavior, allowing for the creation of large amounts of synthetic data by simulating various scenarios and conditions. But synthetic data are limited, as they often do not replicate nuances of real-world data. Tools that enable the integration of real-world, expert labeled data with synthetic data will solve some of these accuracy issues. In addition, vendors should look to support customers that want to train models with multi-modal data as this will improve model performance (generation capabilities and understanding).

5.3. MULTI-MODAL LVMS

LVMs are the natural successor to basic CV, but it is important not to ignore VLMs. VLMs are more complex and integrate the capabilities of Large Language Models (LLMs) with LVMs making them multi-modal. Static VLMs that can analyze images and provide text responses are already in operation today, but more complex, dynamic use cases for other applications are emerging: image captioning in industrial—providing text feedback on incoming image streams; visual question answering—enabling employees to query videos to understand problems and faults; autonomous vehicle drive assist—enabling drivers to understand Advanced Driver-Assistance Systems (ADAS) decisions; and healthcare diagnostics—capable of analyzing images and providing feedback.

6. EXPLORING THE FUTURE OF VISION AI

6.1. EVALUATING CV AND LVM VALUE PROPOSITIONS

From a purely technological perspective, LVMs surpass the capabilities of “traditional” CV models by being able to handle more complex tasks with better generalization and domain adaptability. But this is not the full story, so Table 3 provides an assessment of each model’s value proposition.

Table 3: Comparing CV and LVMs

(Source: ABI Research)

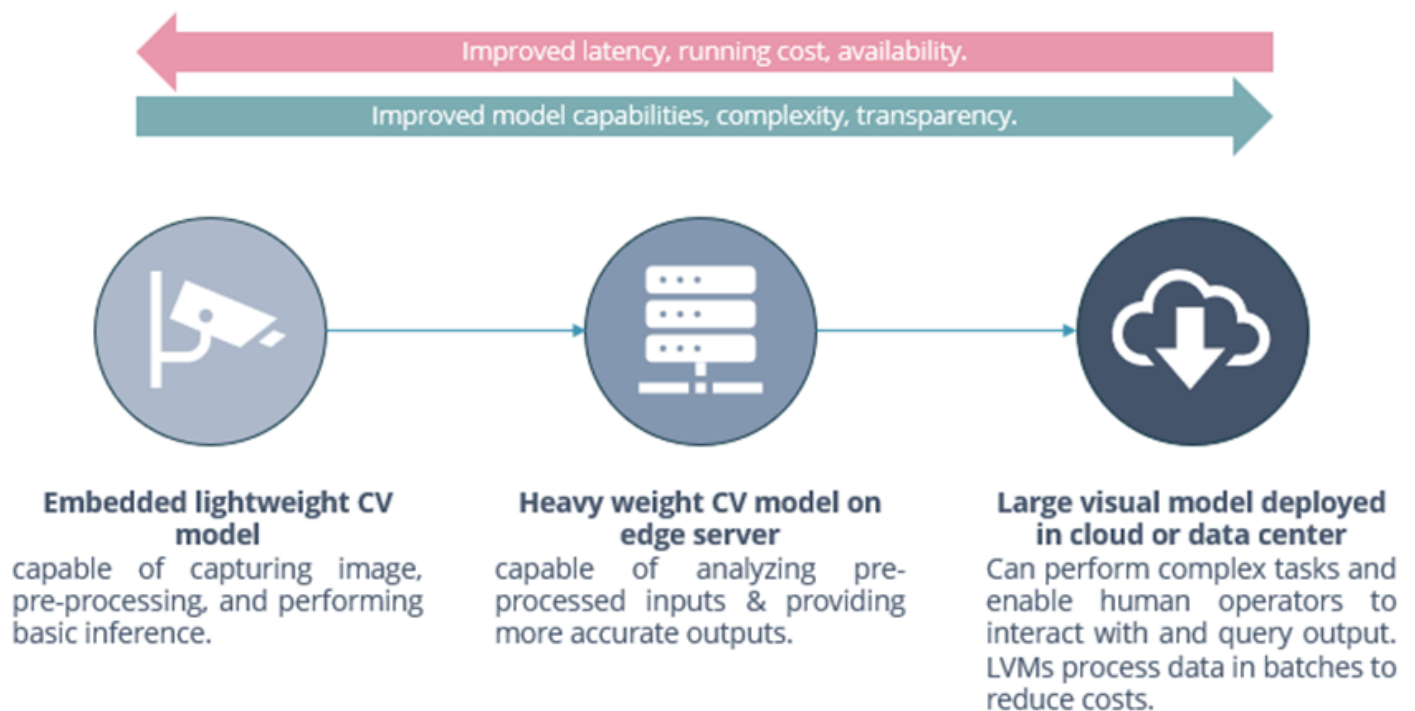
Factors	Computer Vision	LVMs
Processing Latency	Low: Can support 15 Milliseconds (ms) latency for real-time use cases.	High: Cannot support most real-time use cases.
Resource Requirements	Low: Can be run with limited compute resources (Central Processing Units (CPUs)).	High: Requires leading accelerators like Jetson to support with sufficient latency.
Total Cost of Ownership (TCO)	Medium: Some Capital Expenditure (CAPEX) to buy cameras, but low Operational Expenditure (OPEX) for model operations.	High: Data processes, cost of training, hardware, implementation, management.
Domain Adaptability	Low: Capable of supporting specific use cases.	High: capable of supporting a diverse range of applications, rather than limited to one.
Time-to-Value	High: Data labeling, supervised training.	Medium: Unsupervised learning with unstructured data.
Data Requirements	Low Volume of Labeled Data	High Volume of Unstructured Data: Requires training on vast data sets to instill effective behaviors.
Risk	Limited: Highly-specific application.	High: Hallucinations occur regularly.
Return on Investment (ROI)	Proven, High ROI	Unknown ROI

6.2. FUTURE OF THE VISION AI MARKET

The market will very slowly transition from CV models to LVMs, driven by numerous factors—hardware innovation, use case development, model R&D, performance of lightweight LVMs, and maturity of related technologies. However, the first stage of this transition will be hybrid AI deployments.

Figure 1: Hybrid AI Implementation

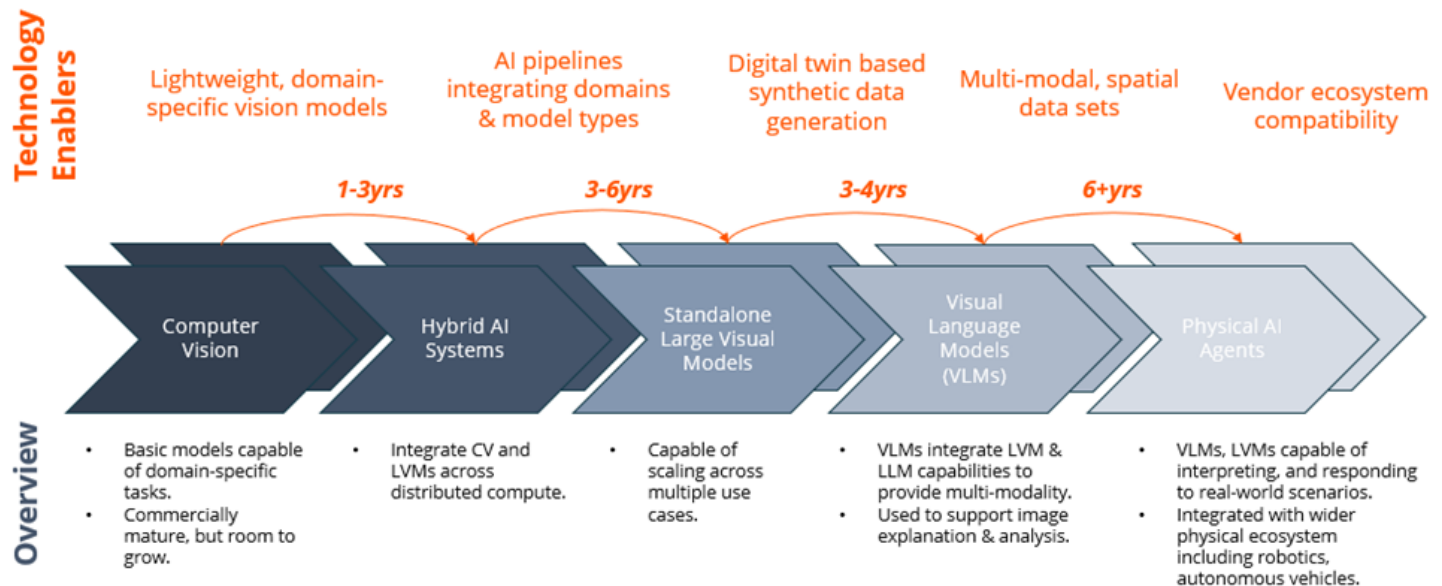
(Source: ABI Research)



Hybrid AI systems and standalone LVMs will not be the end of market innovation. As technology constraints are removed and demand grows for new and more innovative use cases, ABI Research expects further development in the market, as suggested in Figure 2.

Figure 2: Vision AI Deployment

(Source: ABI Research)



7. COMMERCIAL OPPORTUNITIES & CHALLENGES WITHIN THE VISION AI MARKET

7.1. OPPORTUNITIES

- Growing Commercial with Channel Partner Presence from Global System Integrators (GSIs), ISVs: CV ecosystem is continuing to develop with GSIs playing a significant role in implementation.
- Domain-Specific LVLMs Will Start to Gain Traction: Although constrained, the LVM market is showing signs of life with domain-specific deployments.
- Open-Source Market Is Likely to Dominate, Which Will Reduce R&D Costs: Similar to the CV market, LVLMs are showing early signs that open-source will be a major driver. New models are being open-sourced, which will lower barriers for vendors. This will mean that pivoting from CV to LVLMs and beyond will be less expensive than many expect.

7.2. CHALLENGES

- Hybrid AI Systems Are Necessary, but Difficult to Implement: Enterprise interest in hybrid AI is challenging and costly to implement. Building pipelines to support integration will be necessary to enable this deployment with minimal development understanding.

- **LVM Ecosystem Still Not Defined:** Model development remains in a relatively early stage; traditional channel partners like SIs or ISVs are still focusing on the CV segment as plenty of commercial growth is still expected.
- **Enterprises Struggle to Scale Vision AI Deployment:** Many enterprises have individual deployments of vision AI use cases performing specific functions like object detection. However, people, processes, and technologies all impact their ability to scale deployments across multiple use cases. This limits the commercial opportunity for the ecosystem.

Related Research	
Whitepaper	5G And AI: The Foundations For The Next Societal And Business Leap
MD-AIML-108	Artificial Intelligence and Machine Learning
AN-5334	The Edge AI Ecosystem



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157 Columbus Avenue 4th Floor
New York, New York 10023 USA
Tel: +1 516-624-2500
www.abiresearch.com

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